



Guideline

Methotrexate (Oral)

Scope

Site	Service/Department/Unit	Disciplines
SCGH	All Areas	Medical, Nursing, Pharmacy

Introduction

This guideline provides guidance to all Sir Charles Gairdner Hospital (SCGH) staff involved in the prescription, administration, supply and management of patients on methotrexate. This document supports and complies with related legislation, policy and other related documents [here](#).

HIGH RISK MEDICINE

1.0 General Information

Methotrexate is a cytotoxic, immunosuppressant medication usually prescribed by the specialist areas of oncology, haematology, dermatology, immunology and rheumatology.¹

Low dose oral methotrexate (i.e. less than 25mg taken as a single dose once a week) is a safe and effective medication if taken at the right dose with appropriate monitoring.⁴

Prescribing, dispensing and administration errors have led to inappropriate methotrexate dosing, resulting in adverse outcomes including death.¹

Common problems with methotrexate:^{1,2}

- Once weekly dose regimen is uncommon
- The medication comes in a large quantity and can result in a loss of dose warnings or prompts when re-packaging by pharmacists occurs.
- Similar appearance of methotrexate tablets, folic acid 5mg tablets and prednisolone 5mg tablets, often prescribed concurrently, can also result in identification incidents.

Methotrexate is a high-risk medicine because when not used correctly, it is associated with an increased risk of toxicity.⁴

2.0 Presentation⁶

- Methotrexate 2.5mg tablets (Methoblastin®)
- Methotrexate 10mg tablets (Methoblastin®)

3.0 Indications

Used for its immunomodulatory effects in a wide range of conditions including:

- Rheumatoid arthritis
- Psoriasis
- Juvenile arthritis

- Crohn's disease.

Note: Methotrexate is also indicated in the treatment of neoplastic diseases. This guideline does not outline use in these circumstances. Please consult specialist protocols.

4.0 Contraindications⁸

- Infection: serious or untreated e.g. sepsis, hepatitis B, active TB (before completing TB treatment)
- Renal disease where creatinine clearance <10mL/min
- Chronic hepatic disease
- Pregnancy and Breastfeeding

5.0 Precautions

- Infection: may reactivate inactive hepatitis B and latent TB (begin TB treatment before commencing methotrexate)

6.0 Drug Interactions⁸

- There are a number of important drug interactions with methotrexate, consult [SCGOPHCG approved references](#) (e.g. Australian Medicines Handbook) for more information
- Methotrexate is a folate antagonist; combination with other folate antagonists increases the risk of myelosuppression.
- It is also a substrate of organic anion transporting polypeptide (OATP) 1B1; giving it with inhibitors of this protein may increase its concentration and the risk of adverse effects.
- Combination with drugs that affect platelet function, e.g. aspirin or NSAIDs, should be avoided unless the benefit outweighs the risk when using methotrexate in oncology.
- It is also hepatotoxic; treatment with other hepatotoxic drugs, e.g. leflunomide, may increase risk of hepatotoxicity; use such combinations with caution and monitor aminotransferase concentration.

7.0 Prescribing requirements^{1,3,4}

The most common cause of significant patient harm reported in the medical literature from Europe, Australia and US, occurred when a medical practitioner had mistakenly prescribed methotrexate to be taken daily rather than once a week, followed by a pharmacist dispensing the methotrexate accordingly.⁴

A medication review should be undertaken on admission and documented on the Medication History and Management Plan (MHMP).^{1,3,4}

Incident reports relating to erroneous methotrexate prescribing and administration should be used as an educational tool with feedback to prescribers, pharmacy and nursing staff.¹

7.1 Prescribing

When a weekly dose of oral methotrexate is prescribed on the WA Hospital Medication Chart (WA HMC), the prescriber must specify:

- Frequency as "once weekly"
- The day of the week it is administered

- The administration section of the WA HMC must clearly indicate which day the dose is due, with the remainder of administration boxes crossed out to flag dose(s) not to be administered
- Dose in milligrams (mg)
- Route must be specified as oral
- Include the indication on the prescription (this is useful information for all healthcare providers).

Prescribers should query whether patient is on a concurrent folic acid regimen. Folic acid nearly always accompanies methotrexate prescribing and is often forgotten.

The use of 'as directed' on discharge and outpatient prescriptions is unacceptable.

Prescribers should also be aware of drug interactions between methotrexate and any concomitant medications, especially newly initiated medications.⁴

7.2 Discharge of Patients on Methotrexate^{1,2,3,4}

All patients commenced on methotrexate in hospital should be counselled by a pharmacist or doctor and:

- Provide a written information sheet (which should contain ONLY the information applicable to the patient's condition– see Patient Information Sheet from the Rheumatology Association).⁶
- Emphasise the potential toxicity of methotrexate.
- Stress the importance of understanding the weekly dosing regimen and risks of taking a daily dose by mistake
- Explain accompanying folic acid regimen if indicated and ensure this is not confused with methotrexate.
- Provide detailed information about how to use methotrexate safely including issues such as minimising alcohol and caffeine intake, seeking medical attention at the first signs of being unwell and wearing sunscreen
- Supply a medication list, monitoring plan and provide suggestions for remembering weekly administration.

Discharge summaries must be complete and accurate. The discharge prescription must include the full form, strength, dose, and frequency/direction.

On discharge, the patient should also be supplied with a record of laboratory results and be made aware of any ongoing monitoring schedule and signs/symptoms of adverse reactions and toxicity.

A lack of understanding (or misunderstanding) by patients and healthcare professionals of methotrexate and the dosing schedule is a contributing factor in adverse events associated with methotrexate.²

If methotrexate is supplied on discharge as part of a SCGOPHCG Webster Pack, it will be packed separately and labelled clearly as Cytotoxic.

8.0 Dose⁸

Methotrexate is taken **once a week** (on the same day each week)

When changing from oral to SC/IM: consider reducing the dose and/or monitoring carefully, especially for oral doses >15 mg, as bioavailability is higher with parenteral administration.

Renal impairment

Adult, CrCl 10–50 mL/minute, reduce initial dose by up to a half; monitor and adjust dose as needed.

9.0 Administration^{1,3}

Any questions or confusion regarding methotrexate orders must be referred to the prescriber or pharmacist for clarification prior to administration.

All prescriptions should be double checked before administration. The WA HMC should be checked by the pharmacist prior to administration of methotrexate to ensure the day the dose is due is clearly indicated and the remainder of administration boxes are crossed out.

Staff should administer and handle this medication in accordance with the [Nursing Practice Guideline No. 51 – Medication Management](#) and [Nursing Practice Guideline No. 52 – Antineoplastic/Hazardous Agent Management \(in the cancer and non-cancer setting\)](#).

Safe work procedures stress the need to:

- Wear personal protective equipment such as gloves.
- Non-touch technique should be used for the administration of oral cytotoxic medications to avoid direct contact with the tablets, such as disposable cups.
- Tablets should never be crushed, broken or split, due to the risk of inhalation by any persons in the immediate area. If lower doses or enteral tube administration (e.g. nasogastric) are required, contact the ward pharmacist.
- Any concerns related to oral methotrexate should be referred to the ward pharmacist or pharmacy department.

*ALL employees should be fully informed of the potential, but yet unproven, reproductive hazards of cytotoxic medications and, if they so request, employees who **are pregnant, breastfeeding or actively trying to conceive should not be involved in handling of cytotoxic medications.***

9.1 Storage and Supply⁴

- Measures are taken to reduce errors associated with the storage of different strengths of methotrexate tablets in the Pharmacy Department. Warning signs and shelf labelling are utilised to minimise the risk of the incorrect strength being selected.
- Methotrexate is not available through imprest in ward areas and will need to be ordered through the clinical pharmacist during working hours. After hours supply is unlikely to be necessary and increases the risk of medication misadventure. If deemed clinically appropriate by the medical team and the after-hours CNS the on-call pharmacist will need to be contacted.
- Each dose of methotrexate should be reconciled by a pharmacist before administration. The pharmacist will order each dose from pharmacy (exact quantity of tablets to make up one dose). Each dose will be supplied on the day of administration and will be labelled with the patient's name, the dose, the date and the day of the week that the dose is due. Unused or unwanted supplies are to be returned to pharmacy as soon as practicable.

- It is advised for any pharmacist dispensing this medication to confirm the dose with the Clinical Pharmacist or prescribing Doctor should they have any concerns or uncertainties with the order.
- Methotrexate should be clearly labelled with a cytotoxic warning label and a special handling required label.
- Methotrexate must always be supplied with a child-resistant cap.

10.0 Monitoring^{3,4,6,8}

The treating specialist must specify an individual monitoring plan for the patient. Laboratory monitoring should be done at baseline and then repeated at regular intervals until the patient is stabilised and then on an ongoing basis so that the patient can be evaluated to prevent methotrexate toxicity.

Note; Increased monitoring may also be necessary with large dose increases and in patients with co-morbidities.

X-rays and respiratory function tests should be considered prior to starting treatment with methotrexate if they have not been done in the previous 12 months.^{3,4}

Laboratory monitoring will include:

- full blood count
- renal function
- liver function tests
- hepatitis B & C serology (in high-risk patients).

11.0 Adverse Effects^{4,8,10}

The most common side effects are gastrointestinal symptoms (stomatitis, nausea, vomiting, diarrhoea).

All patients must be advised to report symptoms of adverse reactions to their treating physician or GP.

Patients are more at risk of methotrexate toxicity if they are prescribed medications causing pharmacokinetic interactions, have renal dysfunction or are elderly.

TOXICITY

Methotrexate may cause significant toxicity which can occur at therapeutic dosing.

Significant toxic effects can present as:

- Vomiting and diarrhoea
- Bone marrow suppression (fever, sore throat, mouth ulcers)
- Hepatotoxicity (abdominal pain, jaundice)
- Pulmonary toxicity (new or increasing dyspnoea, cough, chest pain, hypoxaemia)

Refer to Toxicology & Haematology Departments for further information on treatment of oral methotrexate toxicity and overdose. Antidote therapy includes treatment with CALCIUM FOLINATE - see [Therapeutic Guidelines](#) for dosing information.

12.0 Pregnancy & Breastfeeding:^{9,10}

- Contraindicated in Pregnancy and breastfeeding

- Contact the Obstetric Medicines Information Service, phone: 6458 2723

Related Documents

Legislation:

- [Medicines and Poisons Act 2014](#)
- [Medicines and Poisons Regulations 2016](#)
- [Health Services Act 2016](#)

Authorising policy and standards

- [SCGOPHCG Medicines Management Policy and Procedure](#)
- [SCGOPHCG High Risk Medication Policy](#)
- [Mandatory Policy MP0077/18 Statewide Medicines Formulary Policy](#)
- [OD 0647/16 National Standard for User-Applied Labelling of Injectable Medicines, Fluids and Lines 2016](#)
- [Australian Commission on Safety and Quality in Healthcare. Recommendation for terminology, abbreviations and symbols used in medicines documentation, December 2016](#)
- NSQHS Standards: 1,4,6

Supporting documents

- Australian Commission on Safety and Quality in Healthcare. National terminology, abbreviations and symbols to be used in the prescribing and administering of medicines in Australian hospitals. 2008.
- [Nursing Practice Guideline No. 51 – Medication Management](#)
- [Nursing Practice Guideline No. 52 – Antineoplastic/Hazardous Agent Management \(in the cancer and non-cancer setting\).](#)
- [Medicines Management](#) SCGH Hospital Policy 246. Reviewed 2023.
- [Guidelines for the WA Hospital Medication Chart \(Adult and Paediatric\)](#)
- [SCGOPHCG High Risk Medication Policy, 2023](#)

References

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For full details regarding, development, review, consultation, approval, endorsement - contact the Policy Officer for the submission form

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